

Correlation Visualization for Multivariate Volumes with Continuous Indexed Points

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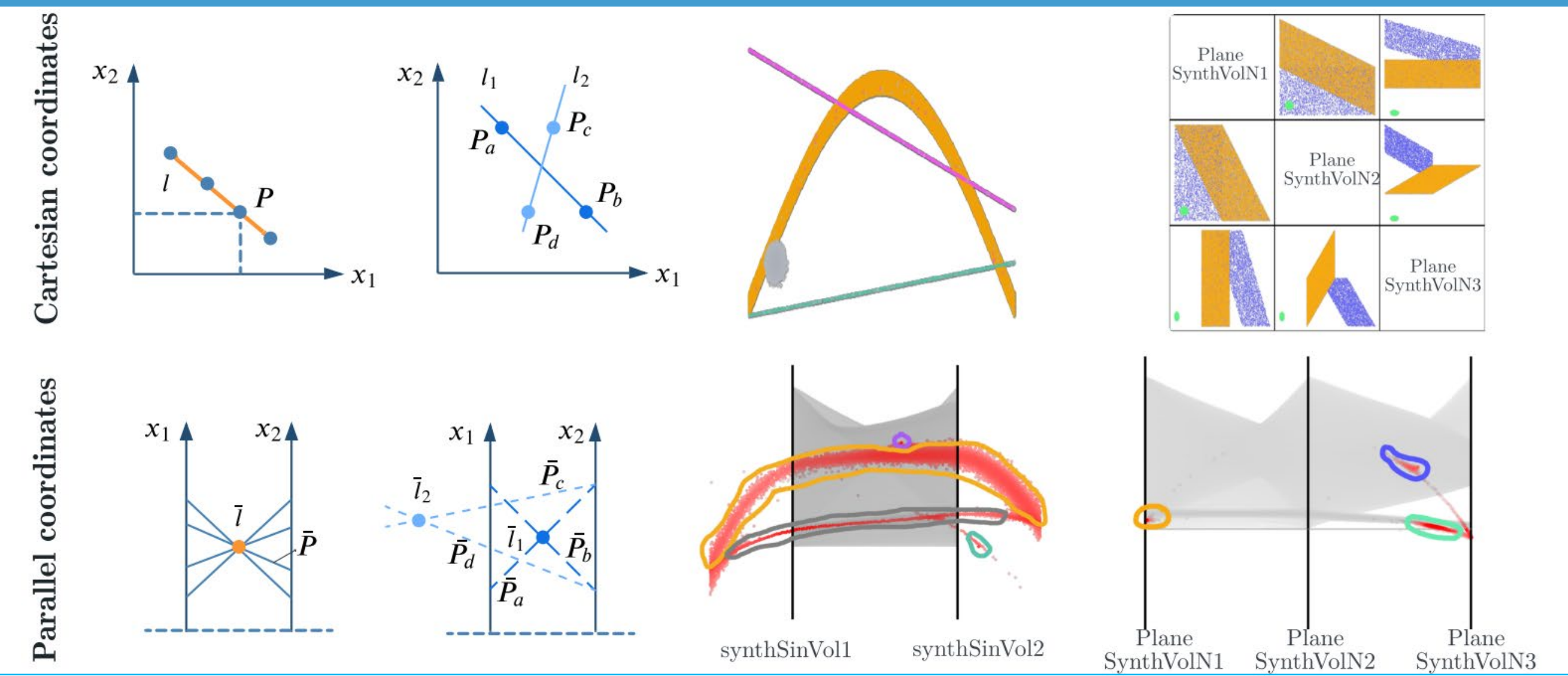
北京大学 健康医疗大数据
国家研究院
NATIONAL INSTITUTE OF HEALTH
DATA SCIENCE AT PEKING UNIVERSITY



Contributions

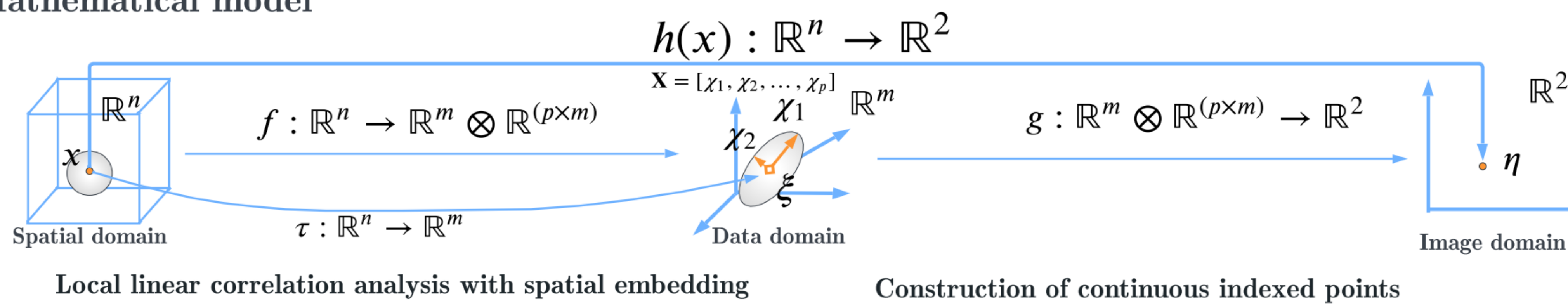
- A representation of local correlations in spatial neighborhoods of volumetric data
- Continuous indexed points as a density model of correlations in parallel coordinates
- Multivariate volume rendering with an advanced shading model

Correlation Representation with Indexed Points



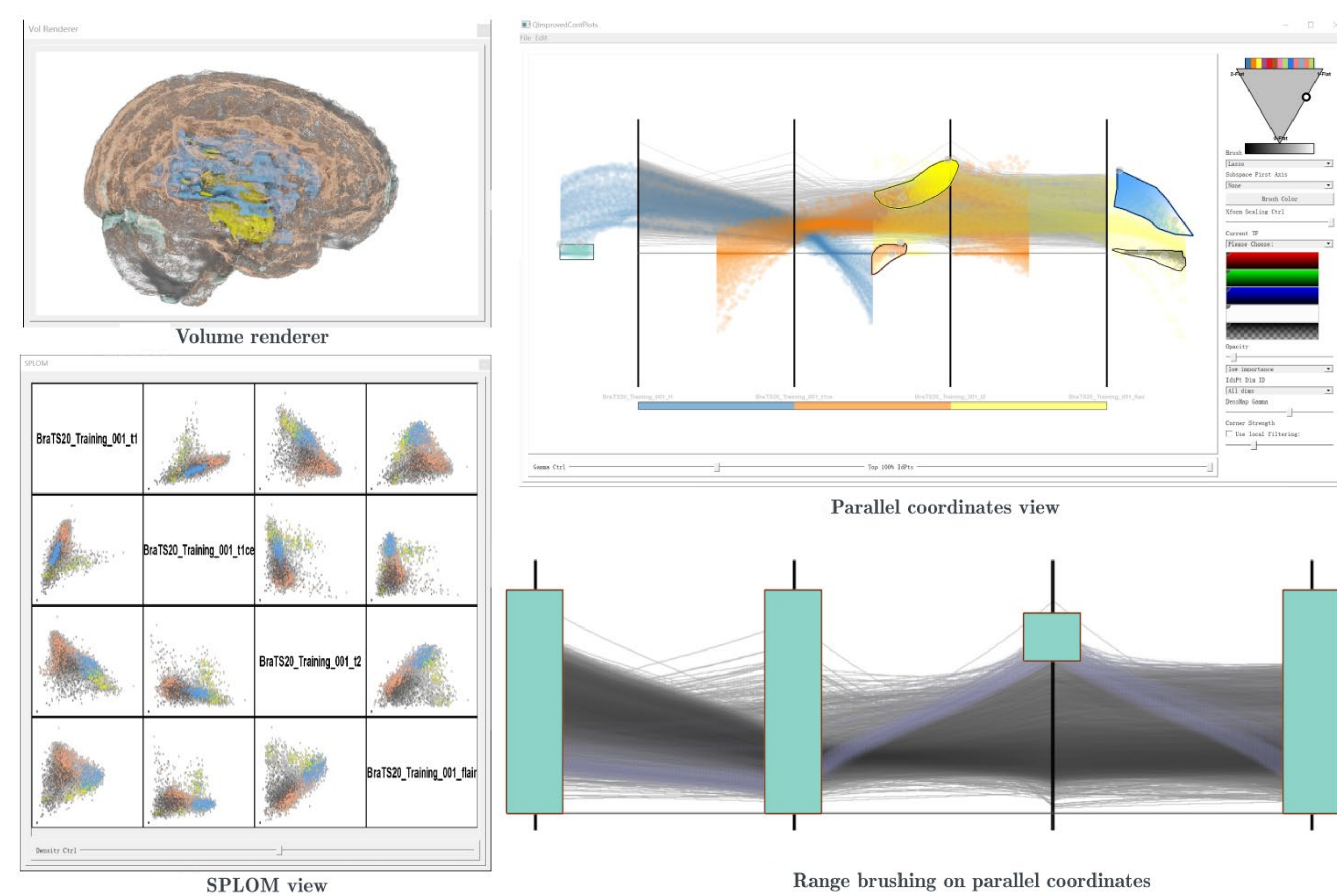
Computational Workflow

Mathematical model



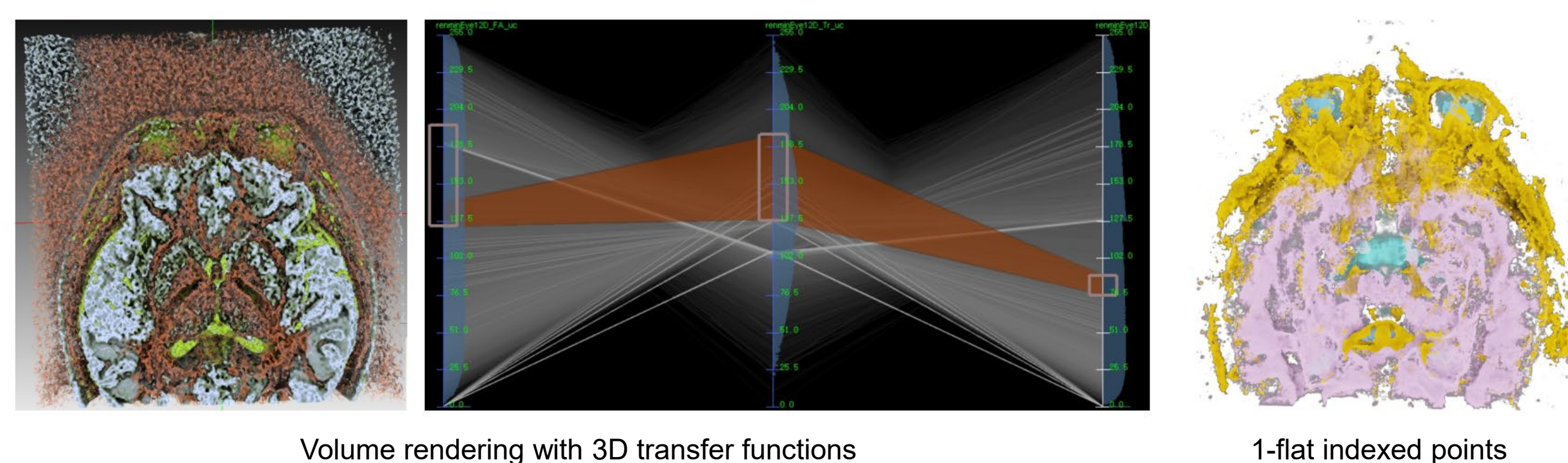
- Hierarchical spatial structure construction
- Local linear fitting
- Indexed points computation
- Angle-uniform transformation
- Density estimation of continuous indexed points

Interactive Visualization



- Continuous indexed points rendering
- Multivariate volume rendering with directional occlusion shading
- Multivariate transfer function widgets
- Brushing-and-linking

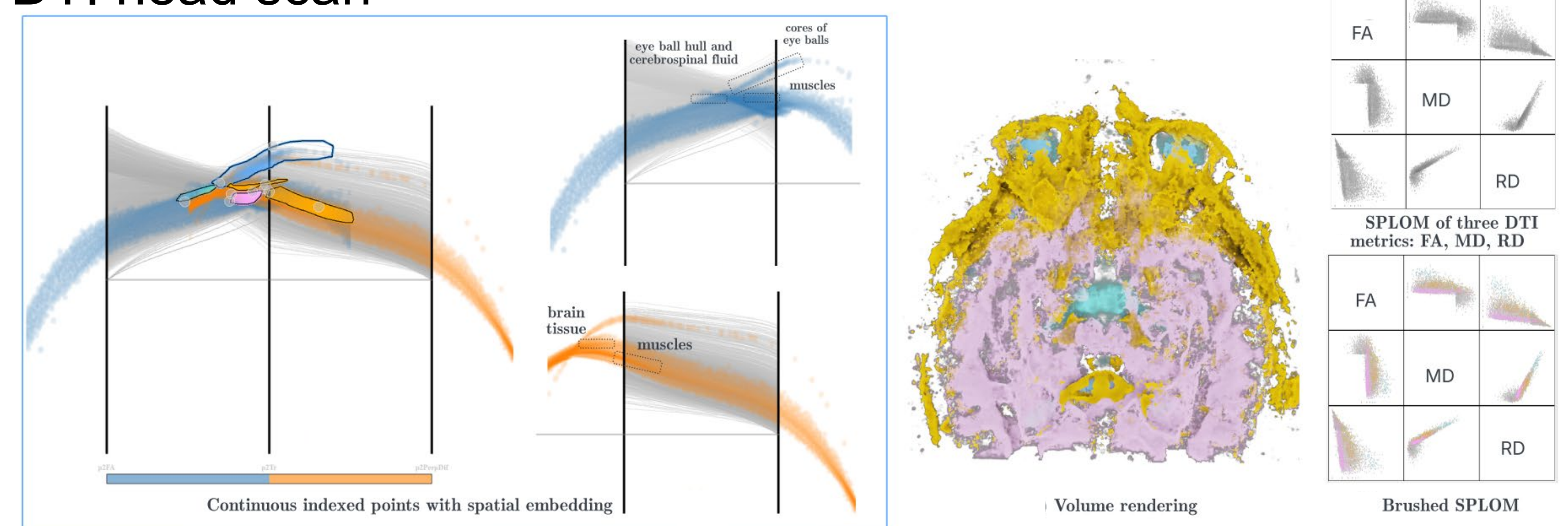
Evaluations



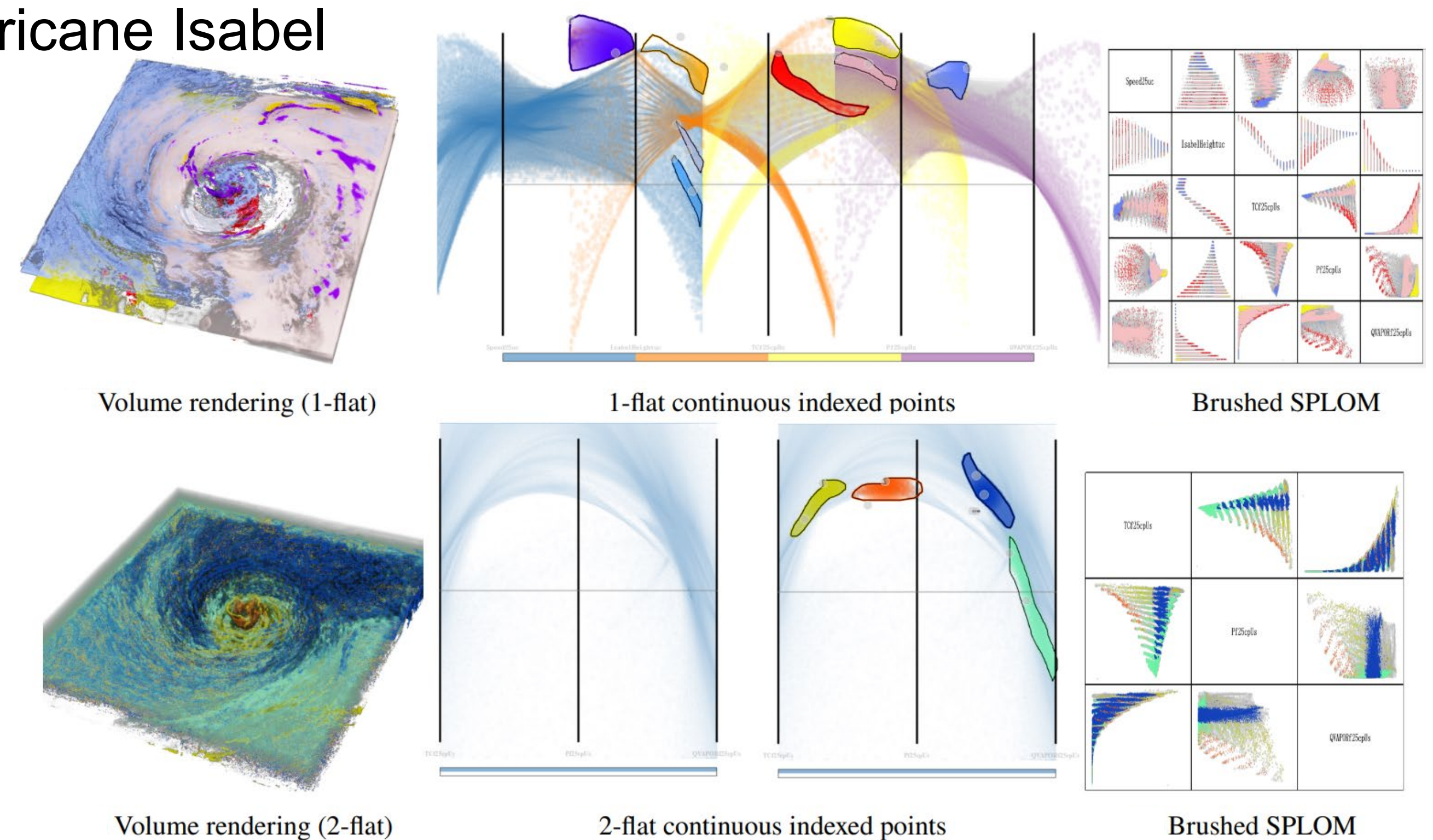
- Comparison to traditional transfer function methods
- User study with visualization experts
- Domain expert feedback

Results

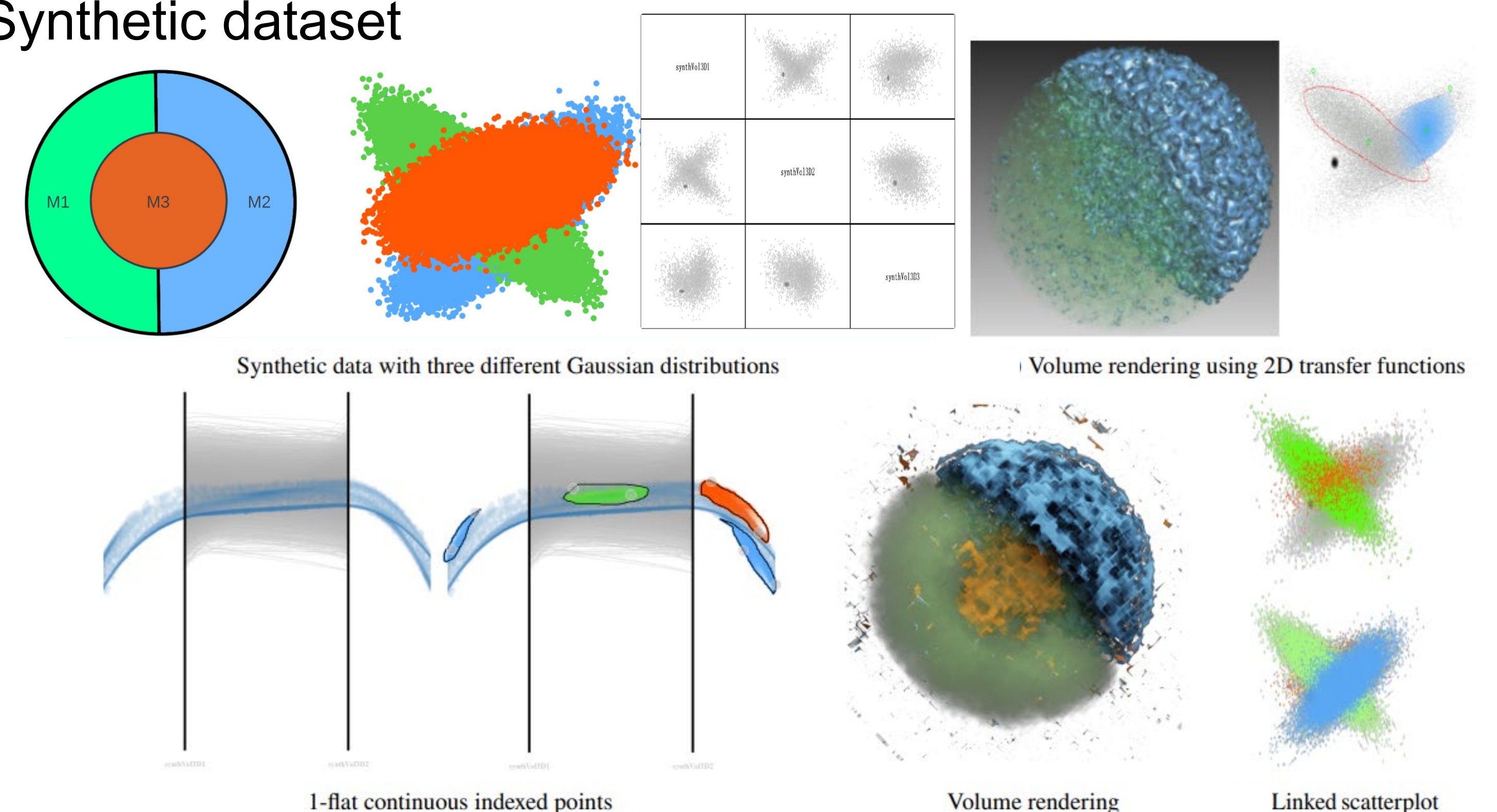
DTI head scan



Hurricane Isabel



Synthetic dataset



Availability

- Publication: L. Zhou, X. Gou, and D. Weiskopf. Continuous indexed points for multivariate volume visualization. Computational Visual Media, in press. DOI: 10.26599/CVM.2025.9450496.
- Authors' version: <https://arxiv.org/pdf/2506.19400>. Code and tool: <https://github.com/zhoul/ContIndexPtVolRen/>



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